# Session Objectives

1) Students will apply the pathophysiology of seizures to distinguish between different types of seizure presentations.

2) Discover how genetic mutations can cause epilepsy syndromes.

3) The students will be informed about different types of epilepsy therapy approaches.

# Seizures Throughout History (Not Tested)

- The word "seizure"  originates from the Latin word sacire, meaning "to take possession of" or "take by force".  The word seizure" was first used in the Middle English period (1150—1500).
- The word "epilepsy" comes from the Greek word epilēpsía, which comes from the verb lambánein, meaning "to seize". The word epilambanein means "to seize, possess, or afflict".
- Seizures and epilepsy in humans and the concept in general has been documented as early as 2500 BC. In the Homeric Era, people believed that epilepsy was a sacred disease induced by gods. In his work " On the Sacred Disease", Hippocrates challenged this perception of a goldy origin of seizures. He proposing that seizures and epilepsy had natural or biologic causes.

- The pt undergoing trepanation (drilling into skull) and cauterization at simultaneously.
- Seizures have been apart of human culture and experience for thousands of years. A lot philosophy behind medical practice was superstition. Trepanation → to drain evil spirits
- People have been making noble yet rudimentary steps at trying to treat seizures and epilepsy for hundreds of years. Painting depicts a surgical approach.
- Careers in neurology with epileptology

- Epilepticus Sic Curabitur ("the way to cure an epilepticus") From Sloane Manuscript, collection of medical manuscripts, end of the 12th century. Author unknown. One of the first artistic descriptions of epilepsy surgery

**Seizure**: – **clinical sign**

- Transient alteration in CNS sensory, motor, autonomic, and/or cognitive function due to excessive & synchronous abnormal neuronal activity from a population of cells in the brain (neocortex, thalamus, etc).
- This excessive & synchronous abnormal neuronal activity leads to disruption in the homeostatic balance between excitation and inhibition.
- **Can have seizures but not have epilepsy, but cannot have epilepsy without seizures→ strict definition later slide**

Drugs, chemicals, compounds, etc reported to induce seizures

| | | |
|---|---|---|
| Alcohols and glycols | Cardiovascular agents | Mushrooms |
| Anesthetics, general | Drugs of abuse | Neuroleptics |
| *Antibiotics | Drug withdrawal | Neuromuscular blockers |
| Anticonvulsants | Hypoglycemics | NSAIDs |
| Anticholinergics | Hydrocarbons | Opioids |
| Antidepressants | Immunosuppressives | *Plants |
| Antifungals | Inhalants | *Radiographic contrast media |
| Antihistamines | Insecticides | Rodenticides |
| Antineoplastics | Metal Chelators | Sympathomimetics |
| Antiparasitics | Insect repellent | Vaccines (due to fever) |
| Antivirals | Metals | |
| Asphyxiants | Muscle relaxants | |

- Don't memorize this

**Causes of seizures are diverse**

- head injury, acute metabolic or electrolyte changes, medications, stroke, tumor, fever, hydrocephaly, infection (neurocysticercosis, COVID-19), etc… )
- Can be **focal** (discretely localized to one hemisphere) or **generalized** (diffusely distributed across both hemispheres.)

**VITAMINS – mnemonic for seizure cause**

**V**ascular event

**I**nfection

**T**rauma

**A**utoimmune

**M**etabolic

**I**ngestion or w**I**thdrawal (ex. Alcohol)

**N**eoplasm

p**S**ych →EEG negative

**P**regnancy → Eclampsia

# More on Terminology/Experiential Phenomena

Convulsion

- Old and vague term typically used to denote a tonic-clonic seizure
- May be used to indicate a seizure with prominent motor activity

Automatism

- Repetitive motor activities that are coordinated and resemble a voluntary movement but are not purposeful (lip-smacking, chewing, swallowing, opening/closing fingers).
- This is not tardive dyskinesia!!! D2 receptor antagonism on cholinergic neurons in the striatum of the basal ganglia leads to aberrant and excessive acetylcholine release causing the features of TD. Pt who are manic, or psychotic will get antipsychotic medications in the ED setting, see on rotations.
- Many classifications of automatisms **(gelastic = bouts of laughing/giggling without other convulsive seizures; dacrystic = bouts of crying with lacrimation. Often follow bouts of gelastic seizure.)**

Sensory/subjective phenomena

- Visual (occipital lobe) and auditory (temporal lobe) hallucinations (Flickering/flashing lights, visual loss, spots in vision or hearing changes
- Somatosensory (paresthesia's, shock-like sensations, numbness, pain, or a sense of movement or a desire to move a body part); Gustatory (e.g. metallic taste)

Experiential phenomena

- Fear, sadness, elation;
- feelings of familiarity (déjà vu) or unfamiliarity (jamais vu);

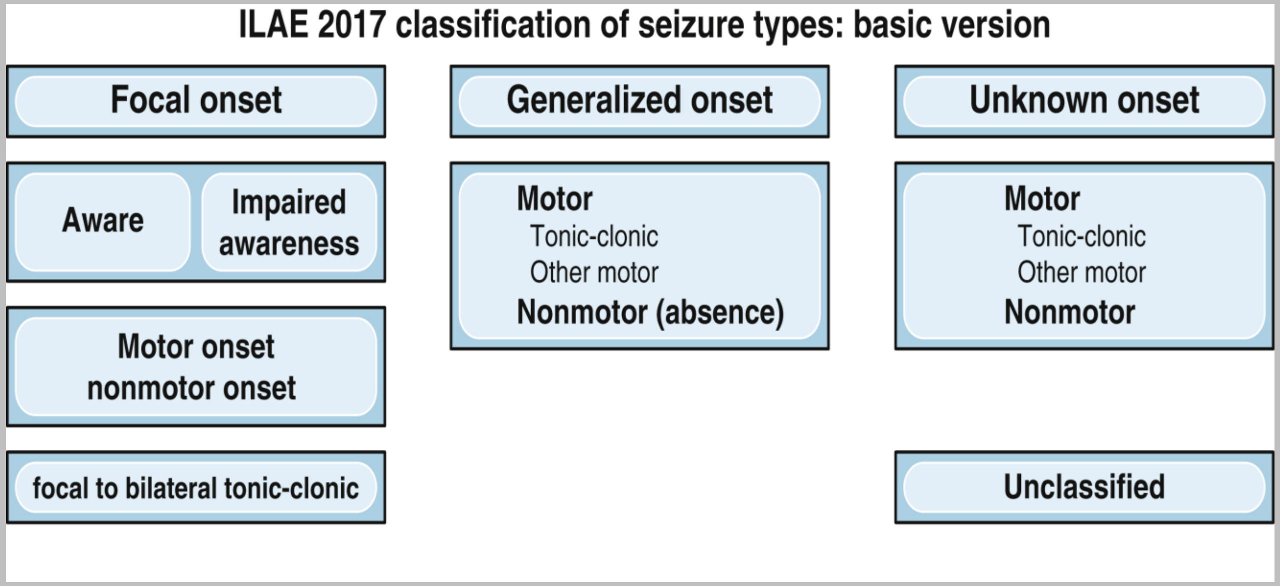
complex hallucination (such as seeing people or hearing music) and illusions (alterations of perception

# Classifying seizures…*always evolving with advances in medicine…*

Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission for Classification and Terminology. Fisher et al. 2017 Epilespia

1] **Generalized seizures** are a result of abnormal neuronal activity **on both sides** of the brain & typically cause **loss of consciousness**, falls, or massive muscle spasms.

2] **Focal seizures** occur in **one hemisphere or one lobe in one hemisphere**. For example, someone might be diagnosed with focal frontal lobe seizures (based on EEG). Partial is old term as well as simple



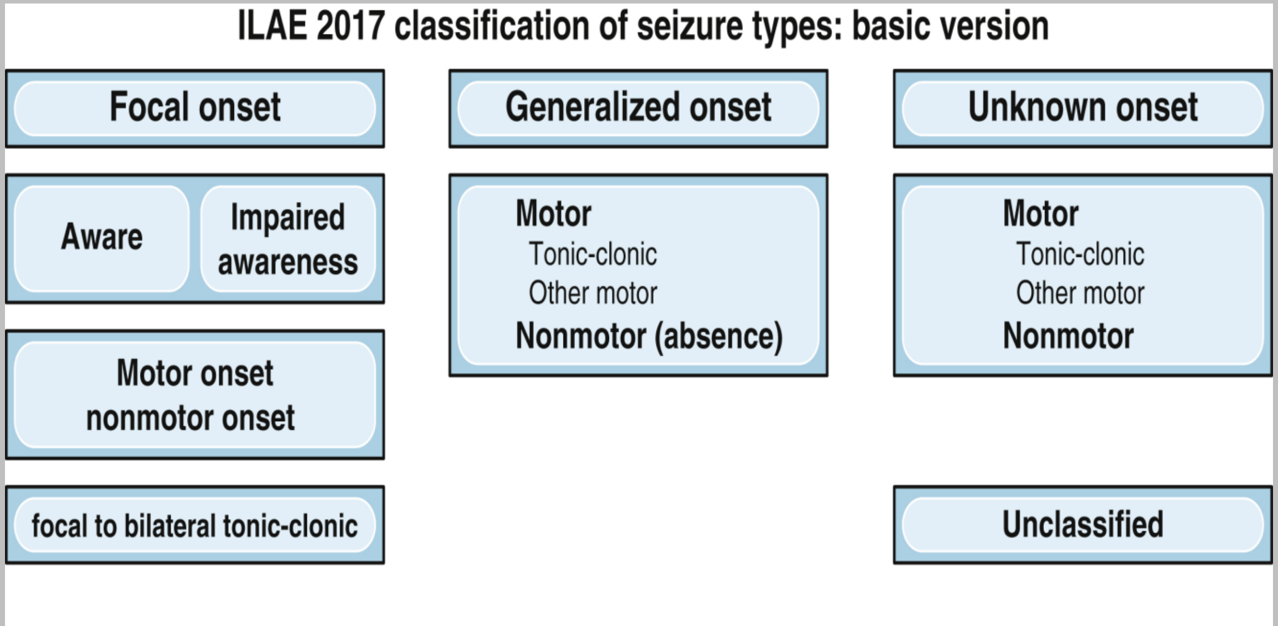

Old terminology

New terminology ( know for exam)

- Simple Partial → Focal aware
- Complex Partial → Focal Impaired awareness

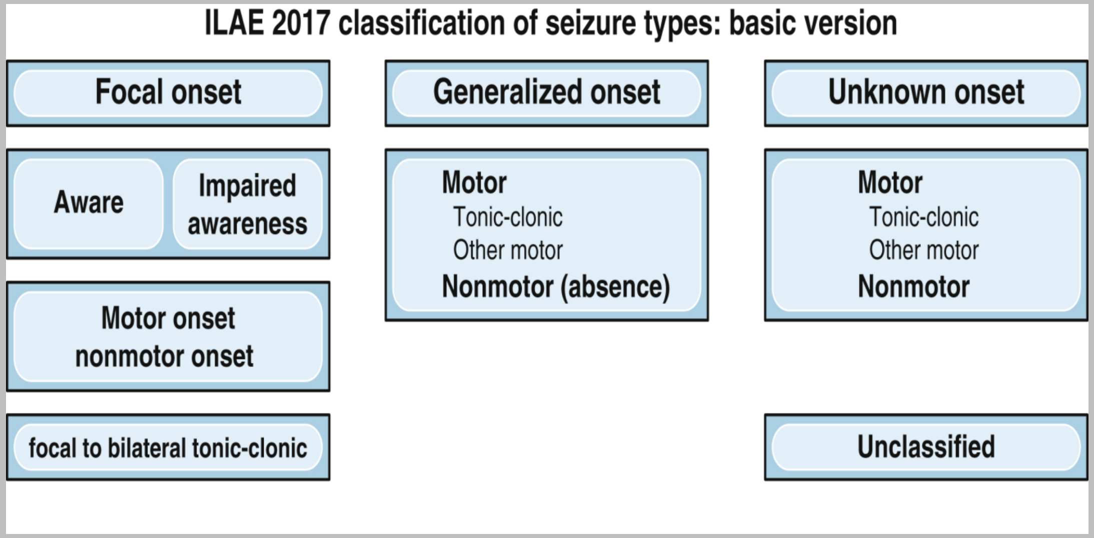
# Focal Onset: Focal Aware Seizures

- Formerly called *Simple Partial seizures*
- *Focal → Originates on one hemisphere of the brain, can become generalized if seizure activity crosses corpus callosum or anterior commissure*
- *Automatisms can occur such as involuntary lip-smacking or blinking*
- **Awareness or mental status is not impaired**
- Focal aware seizures can have auras which are = subjective and experiential phenomena such as alterations of sensory perception (like taste and smell.) The manifestations depend on brain region involved
  - E.g. clonic movement of the left upper extremity indicates right pre-central gyrus seizure activity. Think of the homunculus!!



# Focal Onset: Focal Impaired Awareness Seizure

- "FAIS"
- Formerly known as Complex Partial Seizures (complex = altered mentation)
- Characterized by altered awareness or mental status during the seizure
- May be very subtle, manifesting with slight confusion, or slowing of responses
- May experience amnesia after the event
- Sometimes difficult to determine if awareness was impaired.
  - may be totally conscious (Focal-aware) but unable to respond verbally because of aphasia/motor deficit. Pt may be aware during seizure, but seizure is occurring in the Broca's area, may look like pt cannot follow commands or has alerted mental status
- Most common location is temporal lobe
  - Common in the uncal formation and the hippocampus → can maybe explain the amnesia a person may experience
- Most common motor manifestation is automatisms

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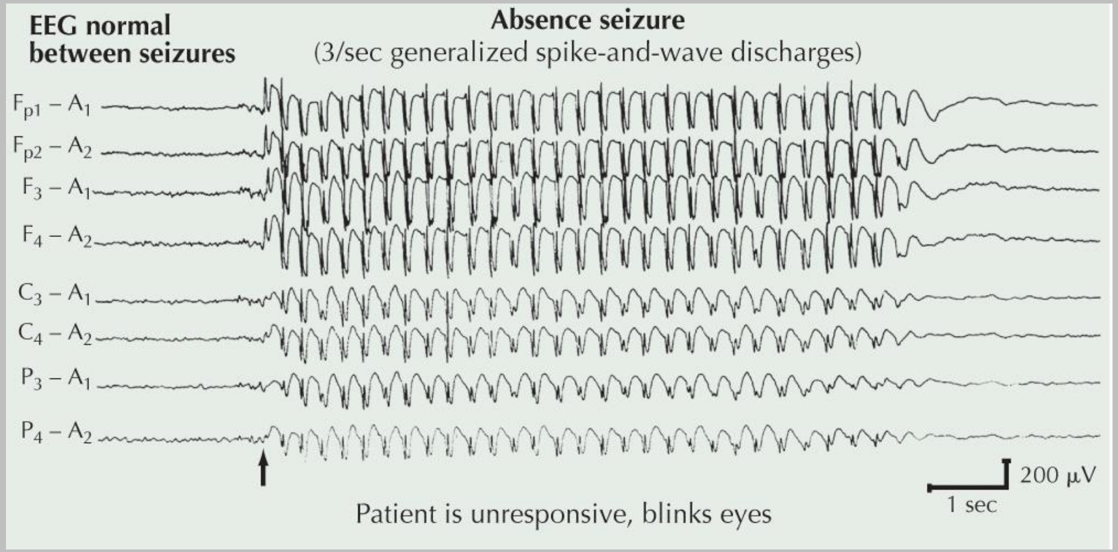
# Generalized Tonic-Clonic Seizure

- **Tonic:** Upward eye deviation with eyes half open and the mouth open. There is involvement of the respiratory muscles usually produces a forced expiration that produces a loud guttural vocalization, often referred to as the epileptic cry.
- Cyanosis may occur during the tonic phase in association with apnea. The tonic phase gradually evolves to clonic activity.
- **Clonus:** Series of rhythmic jerky motions of clonus that slowly decrease in frequency over the course of the seizure
- **Postictal:** The individual is limp and unresponsive.
- Respiration is loud and snoring → as pt regains control over masseter and other jaw muscles. ABC’s !!! Make sure airway is patent. Pt may awaken briefly with postictal confusion.
- Tongue biting common on the side of the tongue→ not a reliable indictor if someone had a seizure.
- Incontinence of urine and stool.

# Generalized Seizures: Absence Seizures

- Sudden blank stare with motor arrest, usually lasting less than 15 seconds
- Usually unresponsive and unaware
- Returns immediately to a baseline level of function with no postictal confusion, important for step exams
- EEG during absence seizure: 3-5 hz spike & dome/wave
- Resemble slow-wave sleep EEG. Behavior during absence seizure resembles a sleep-like state.
- High-yield for boards
- EEG hallmark of a typical generalized absence seizure is generalized 3- to 5-Hz spike-and-wave activity with a normal interictal background.
- Hyperventilation is can be a trigger, can be used for diagnosis




- Clinical presentation may look something like this: A child presents to the pediatrician for complaints of ”staring off into space” or ”not paying attention or focusing in class or at home when told to do things”. If absence seizure is suspected, the physician will ask the child to hyperventilate to see if this will elicit an absence seizure. This will help confirm our diagnoses and this pt will be most likely treated with ethosuximide.
- Treatment: **ethosuximide**, non -selective inhibition of Ca+ T-type channel Cav1,Cav2, Cav3 variants
- Long term prognosis: most outgrow

# Generalized Seizures: Febrile Seizures

- Affects 2%–5% of all children, mostly between 3 months and 6 years
- Most common form of seizure in children
- Benign in themselves in the vast majority, but do increase the risk of other epilepsies later in life (7% by age 25)
- Most febrile seizures are generalized tonic-clonic in presentation.
- May be prolonged to qualify for the definition of febrile status epilepticus.
- Can cause brain injury (especially hippocampus)
- On step 2 boards, if pt has febrile seizure and you are in the ER, treat with antipyretic such as acetaminophen and discharge the pt once vitals are stable and follow up.
- There is evidence for strong genetic influence (likely polygenic)
- Some genetic epilepsy syndromes are known to start with febrile seizures (Especially Dravet syndrome)

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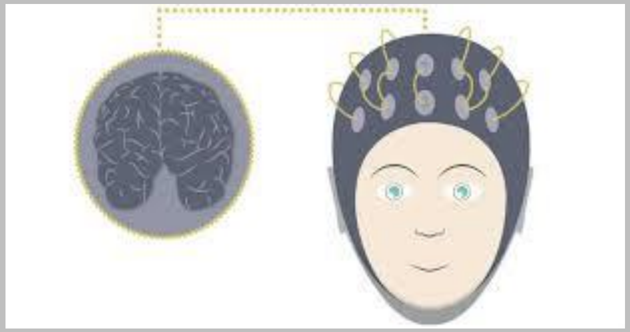
# EEG evaluation – gold standard for seizure detection, done after CT/MRI

Surface EEG



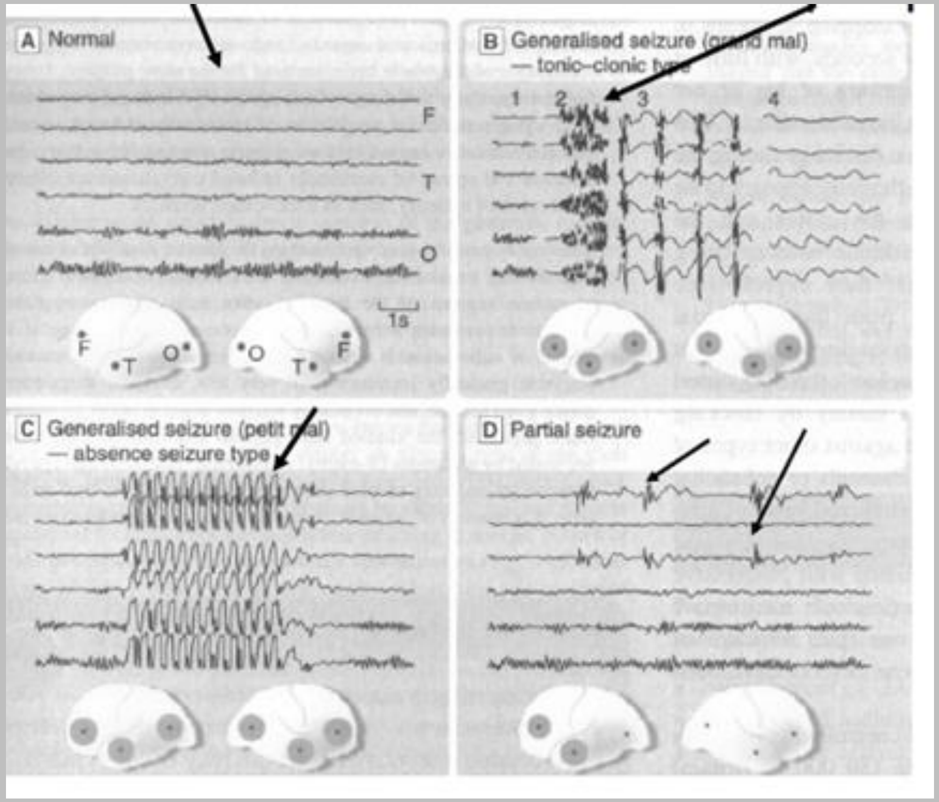
Epidural EEG

- More precision in localizing seizure focus, yet more invasive

- Craniotomy and surface electrodes on surface of brain

Intracranial electrode EEG

Spatial & temporal characterization of abnormal activity.




- CT= very quick, usually ordered before MRI in most emergency settings. CT will be ordered first in order to rule out any structural defects that may be causing the seizure. MRI is more expensive and can help identify more subtle lesions. MRI's take longer to complete than CT scans.

Psuedo-seizures:
* Behavioral events that look like seizures but with no epileptiform activity on EEG.
* Psych manifestation of some underlying issue.
* Treat: **psychotherapy**, though seizures may recur during times of stress

# What is epilepsy?

*Epilepsy:*

*<u>Disease or syndrome</u> in which chronic, recurrent, unprovoked seizures is the salient feature.*

*Diagnosed by <u>extensive clinical work-up</u>*

*i.e. documenting frequency, duration, behavioral profile, etc of seizures,*

*EEG, MR Imaging, bloodwork, PMH, age, family history, genetics, diet, lifestyle, etc…*

Disease of the brain with:

(1) at least two unprovoked seizures >24 hours apart

(2) One unprovoked or reflex seizure in an individual with a high risk of subsequent seizures (e.g., after traumatic brain injury, stroke, CNS infections)

(3) diagnosis of an epilepsy syndrome

- **Seizures and epilepsy are not just about neurons. Mutations can cause defects in astrocytes, endothelial cells and channels associated with these structures. *Think about tripartite synapse**

**65 MILLION**: Number of people around the world who have epilepsy.
**3.4 MILLION**: Number of people in the United States who have epilepsy.
**1 IN 26** people in the United States will develop epilepsy at some point in their lifetime.
**150,000**: Number of new cases of epilepsy in the United States each year
**6 OUT OF 10**: Number of people with epilepsy where the cause is unknown.

Epilepsy Foundation

- About 1 in 100 people in the U.S. has had a single unprovoked seizure or has been diagnosed with epilepsy
- **BETWEEN 4 AND 10 OUT OF 1,000**: Number of people on earth who live with active seizures at any one time.

**Electro-clinical syndromes arranged by age at onset ***

Neonatal period
- Benign familial neonatal seizures (BFNS)
- Early myoclonic encephalopathy (EME)
- Ohtahara syndrome

Infancy
- Migrating partial seizures of infancy
- West syndrome
- Myoclonic epilepsy in infancy (MEI)
- Benign infantile seizures
- Benign familial infantile seizures
- Dravet syndrome
- Myoclonic encephalopathy in nonprogressive disorders

Childhood
- Febrile seizures plus (FS+) (can start in infancy)
- Early onset benign childhood occipital epilepsy (Panayiotopoulos type)
- Epilepsy with myoclonic atonic (previously astatic) seizures
- Benign epilepsy with centrotemporal spikes (BECTS)
- Autosomal-dominant nocturnal frontal lobe epilepsy (ADNFLE)
- Late onset childhood occipital epilepsy (Gastaut type)
- Epilepsy with myoclonic absences
- Lennox-Gastaut syndrome
- Epileptic encephalopathy with continuous spike-and-wave during sleep (CSWS) including: Landau-Kleffner syndrome (LKS)
- Childhood absence epilepsy (CAE)

Adolescence - Adult
- Juvenile absence epilepsy (JAE)
- Juvenile myoclonic epilepsy (JME)
- Epilepsy with generalized tonic-clonic seizures alone
- Progressive myoclonus epilepsies (PME)
- Autosomal dominant partial epilepsy with auditory features (ADPEAF)
- Other familial temporal lobe epilepsies

# Incomplete List of Epilepsy Syndrome

- Don’t memorize this
- All effect ADL and quality of life for patients and family.
- Increased mortality due to  a number of factors.
- All these syndromes have salient features that help distinguish one condition from another differentiate each
- Effects all ages, diverse clinical syndromes that separate each syndrome from the other
- Age of onset matters a lot, adolescent vs neonatal, etc
- Epilepsy fellow will be familiar with these syndromes
- More epilepsy syndromes to come in later slides

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# Causes of epilepsy

These categories are becoming blurred due to increased use of genomic sequencing in clinical medicine.

***1. Structural/metabolic: (MRI/CT positive)***

A distinct condition or disease that has been demonstrated to be associated with a substantially increased risk of developing epilepsy.

- Neuronal migration, brain malformation ex later. Visible on MRI

***2. Genetic: (MRI/CT Negative)***

*a direct result of a known or presumed genetic defect(s) in which seizures are the core symptom of the disorder.*

- **Genetic mutations** encoding for Ach, GABA, Glutamate receptors, Na+, K+, Cl- ion channel. **Not generally visible on MRI**

***3.Unknown/idiopathic***

# Structural/Metabolic causes of both seizures and epilepsy

- Malformations of neocortical development
- Neurocutaneous syndromes (Tuberous sclerosis, Sturge-Weber syndrome)
- Brain Tumor
- Infection (meningitis, AIDS, neuroinflammation)
- Trauma Head injury (25% acute seizures; 10-25% chronic epilepsy)
- Brain Angioma/ vascular malformation
- Peri-natal insult (fetal anoxia)
- Stroke (11% of all new cases of adult-onset epilepsy)

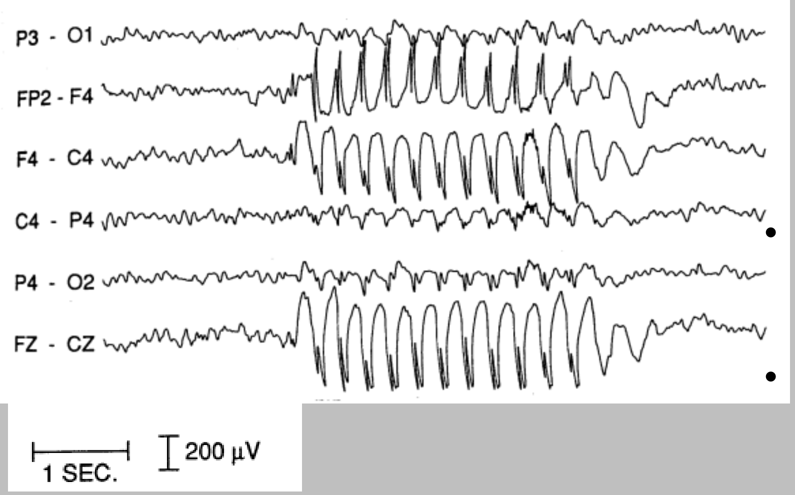
Note: In next slides we will go over a lot of different types of epilepsy and seizures. For your studying and your exam, look for **SALIENT and Distinguishing features** in the descriptions of each disorder. This will help you distinguish one disorder from another.

- Ex) genes, patient presentation, inheritance, age of patient, etc… --> tying these concepts together for each disorder is what Dr. Ramos and myself will test you on

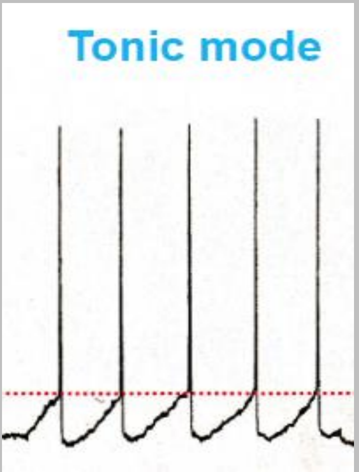
# **Selected Epilepsy Syndromes (Testable on Exam)**: Dravet Syndrome

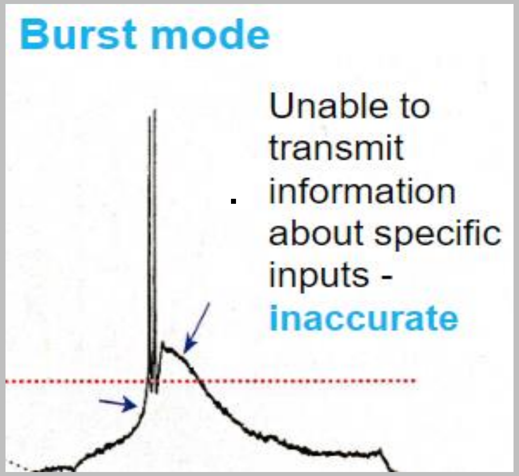
- De novo mutation in **SCN1A gene** - encoding the α-1 sodium channel subunit of voltage-gated sodium channel in neurons.
- **SCN1A gene mutation** has been shown to make Nav 1.1 Na+ channel in neurons open more easily, open for longer periods of time, or re-open faster after closing. More AP's fired!!
- **Good example of epilepsy syndrome with known gene and direct mechanism related to membrane physiology of neurons (action potentials)**
- MRI Positive or **Negative?**
- Also called severe myoclonic epilepsy of infancy
- Typical clinical presentation: previously normally developing infant has febrile status epilepticus at 6 months old, then get recurrent generalized seizures often triggered by fever.
- After 1 year of age, other seizure types appear, including myoclonic seizures, absence seizures, and focal-impaired-awareness and atonic seizures
- The prognosis is poor (intellectual disability, ataxia and spasticity)

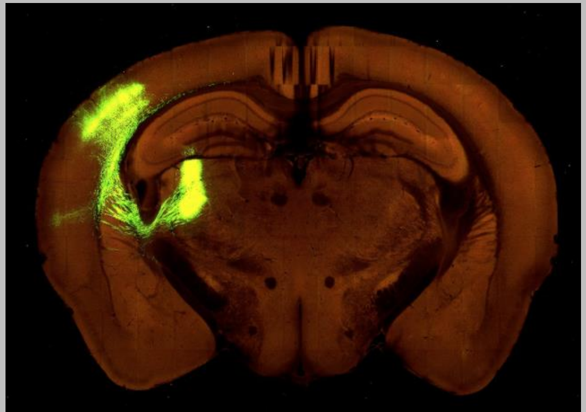
# Cacna1H/ Cacna1A mutations & Absence Epilepsy




- Thalamus has Thalamocortical relay cells (TCR neurons) which are involved in the regulation of sleep-wake states as well as transmission of sensory inputs to the neocortex for conscious appraisal. **When TCR neuron are hyperpolarized (more negative charge inside cell) this cause de-inactivation of Type T Ca2+ channels and eventual creation of burst firing patterns seen in absence epilepsy and SWS**. Gates incoming sensory info to cortex. Tonic mode occurs in (Ach) depolarized TCR neurons (awake and rem sleep)
- TCR neurons have 2 modes, Tonic and burst. Tonic mode occurs in depolarized TCR neurons and this permits consistent signal transmission along TCR neurons. Occurs in REM and wakefulness.
- Excessive calcium influx through the Type T Ca2+ channels in a hyperpolarized TCR neuron is the mechanism from burst mode. This locks the TCR neurons into the characteristic 3 hz spike and dome patterns. This oscillation pattern inhibits transmission of external environmental stimuli.
- The **Cacna1H** mutation thalamic Type T Ca2+ channels of Thalamocortical relay cells cause aberrant calcium influx into TCR neurons during wakefulness of patients with absence epilepsy, causing the EEG finding shown.
- Cacna1a gene may also be implicated in disease process.



- Treatment: **ethosuximide**, non - selective inhibition of Ca+ T-type channel Cav1,Cav2, Cav3 variants
- Long term prognosis: most outgrow

# Lennox Gastaut syndrome

- Lennox-Gastaut Syndrome (LGS) is a lifelong, developmental epileptic encephalopathy
- Onset before age 8, with peak onset between the ages of 3 and 5.
- A recent study estimates the incidence of LGS to be between 14.5 and 28.0 per 100,000 individuals, with a prevalence ranging from 5.8 to 60.8 per 100,000 individuals*
- **Characterized by frequent seizures, including atonic, tonic, and atypical absence seizures → creates neuropsychiatric issues such behavioral disorders and intellectual disabilities.**
- LGS can arise from genetic mutations, cortical malformations, brain tumors, neurocutaneous disorders such as tuberous sclerosis complex, hypoxic-ischemic injury, meningitis, or head trauma. No singular cause. However, in approximately 40% of cases, no identifiable cause is found
- EEG findings may include **interictal 2.5 Hz slow spike-wave discharges and generalized paroxysmal fast activity during sleep**.
- Gene to know: **GABRB3 mutation,** implicated in LGS pathophysiology, makes beta-3 subunit of GABA-A receptor
- Diagnosis relies on clinical presentation, EEG findings, brain magnetic resonance imaging, and genetic testing.
- LGS significantly impacts patients' quality of life, placing considerable strain on caregivers and healthcare systems. Multidisciplinary care needed.

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# Landau-Kleffner Syndrome

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- Landau-Kleffner syndrome (LKS) is a rare age-related epileptic encephalopathy of children **aged roughly 3 to 8 years old with previously normal neuropsychiatric development.**
- Disorder is characterized by **language regression** and seizures and is often associated with social cognitive deficits.
- Disorder characterized by a **developmental regression Broca's and Wernicke's areas, and electroencephalogram (EEG) anomalies located in those areas.**
- **New onset receptive and then later expressive aphasia in previously normally developing individual is hallmark of this disease**
- The seizures, if present, are mainly absence seizures or tonic-clonic episodes which may occur more often during sleep.
- Disorder associated with the **GRIN2A gene**
- This gene makes GluN2A protein: This protein is a subunit of NMDA receptors in neurons.
- NMDA receptors: These are crucial glutamate-gated ion channels in the brain that control the flow of sodium, exciting neurons and transmitting signals.
- Role in the brain: NMDA receptors are known to be involved in learning, memory brain development, and synaptic plasticity.

# Autosomal dominant partial epilepsy with auditory features (ADPEAF)

- Autosomal dominant epilepsy with auditory features (ADEAF) is a focal epilepsy syndrome
- During the seizure, **a patient may have auditory symptoms i.e humming, buzzing, or ringing and/or receptive aphasia**
- The receptive (Wernicke's) aphasia during the seizure consists of a sudden onset of inability to understand spoken or written language without general confusion.
- Other less common symptoms during the seizure may include visual, olfactory, vertiginous, motor, and autonomic symptoms.
- Age at onset is usually in adolescence or early adulthood (age 10-30 years). The clinical course of ADEAF is benign. Seizures are usually well controlled after initiation of medical therapy.
- Since this this disorder has an **<u>AUTOSOMAL DOMINANT</u>** inheritance. Use **Family History** to potentially help you make the diagnosis. If similar presentations have occurred in the patient's family before, this may help you in making your differential diagnosis. Rates of de novo pathogenetic mutations are very low
- Mandatory **criteria to make diagnosis** include a **normal MRI, as well as Focal sensory auditory seizures and/or focal cognitive seizures with receptive aphasia without generalized confusion during the seizure**.
- Genes to know that are implicated in this disorder include *LGI1*, *MICAL1*, or ***RELN*.**
- Know **RELN** for test→ codes for Reelin

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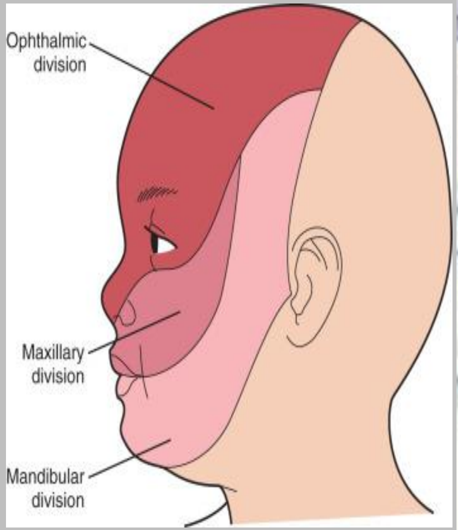
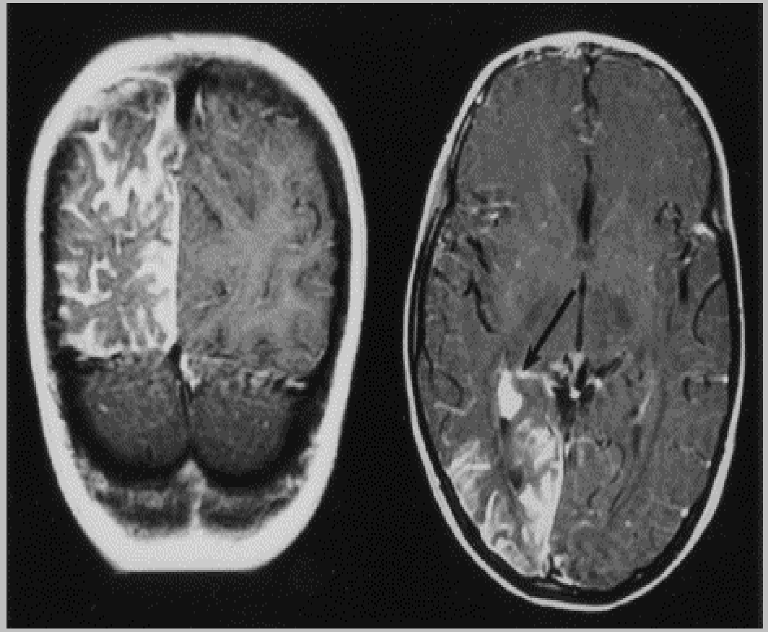
# Sturge-Weber Syndrome

- **Nice example of a disorder with a known gene, robust physical/clinical presentation, characteristic MRI signature**
- Congenital nonhereditary anomaly of neural crest derivatives. **Somatic mosaicism of an activating mutation in one copy of the GNAQ gene. **GNAQ** gene vital for blood vessel development
- Somatic Mosaicism: DNA mutations or alternations in copy number of cellular genomes occur post-fertilization that create two distinct populations of cells. Somatic mosaicism mutations are not inherited, but genetic mosaicism mutations can be.
- Capillary vascular malformations → port- wine stain in Cranial N. V1/V2 distribution; risk of neuro-ocular issues
  - ipsilateral leptomeningeal angioma with calcifications → seizures/ epilepsy; intellectual disability; early-onset glaucoma.



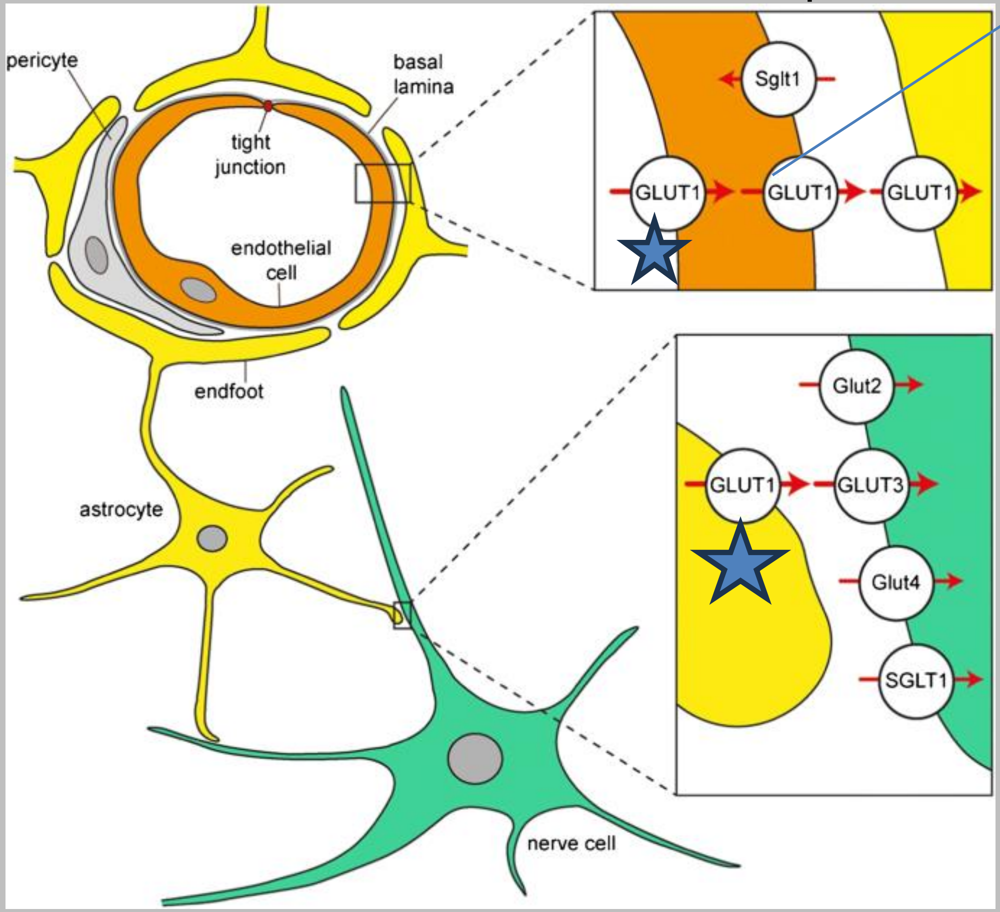




Zitelli & Davis' Atlas of Pediatric Physical Diagnosis, 7th ed.

# Endothelial cells & epilepsy – Glut1 gene (glucose transporter) → ***GLUT1 deficiency syndrome***

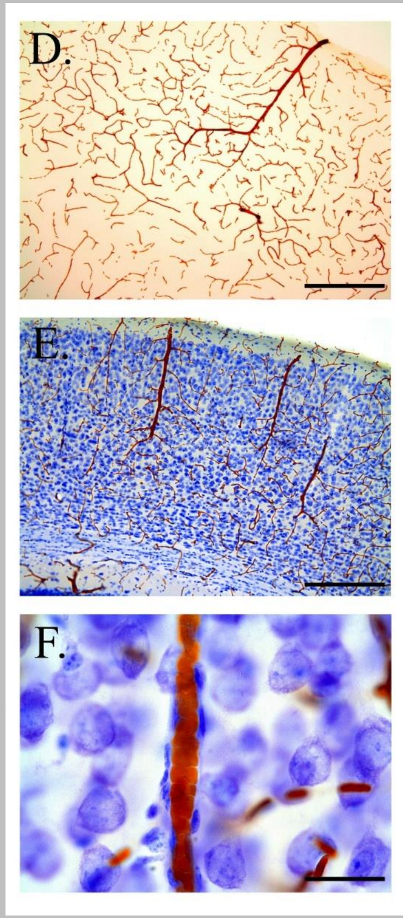
- Astrocytes get glucose from cerebral arteries. Astrocytes transform glucose to lactate and help shuttle lactate to neurons for ATP production




- Glut 1 in brain. Glut 4 mostly in skeletal muscle and adipose

- **Trouble transporting glucose from cerebral arteries into astrocytes**

Ex) after ruling out infectious agent/meningitis and the patient has had a seizure and you see low glucose in the CSF, think about this disorder as a potential cause of the seizure




## ***GLUT1 deficiency syndrome***

- symptoms can include developmental delay, intellectual disability, movement problems, and frequent seizures (diverse types of seizures like atonic, tonic-clonic, etc).
- Problem in vascular transport out of blood into brain
- **Hypoglycorrhachia: low concentration of glucose in the CSF**
- Think of BGL as your 6th vital as a clinician!!!

# Tuberous Sclerosis

Ashleaf spot

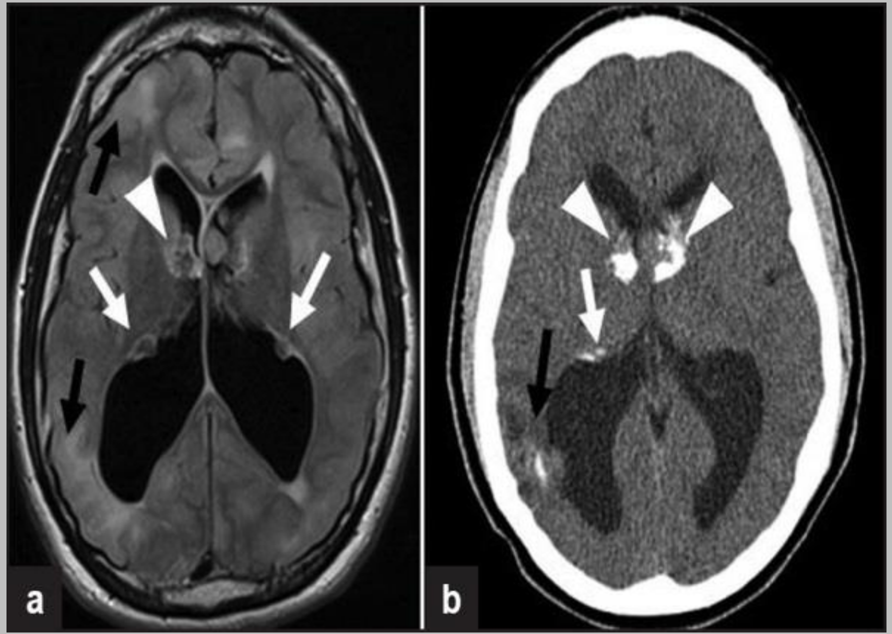
Facial angiofibroma

- Neurocutaneous disorder with **autosomal dominant inheritance pattern** (genes identified in 1990's)
  - **TSC1** chrom 9q → **hamartin**: tumor suppressor (downregulates mTOR)
  - **TSC2** chrom 16p → **tuberin**: tumor suppressor (downregulates mTOR)
  - Genes involved may explain hamartoma presence in various systems
- Clinical mnemonic (**ASHLEAF**)
  - **A**shleaf spots
  - **S**hagreen patches (skin hamartoma)
  - **H**eart rhabdomyosarcoma
  - **L**ung hamartomas
  - **E**pilepsy (brain hamartomas) – difficult to control→ **can cause seizures**
  - **A**ngiomyolipoma in kidney
  - **F**acial angiofibroma

- mTOR=mammalian target of rapamycin. Involved, in cell growth, autophagy and metabolism

Shagreen patch

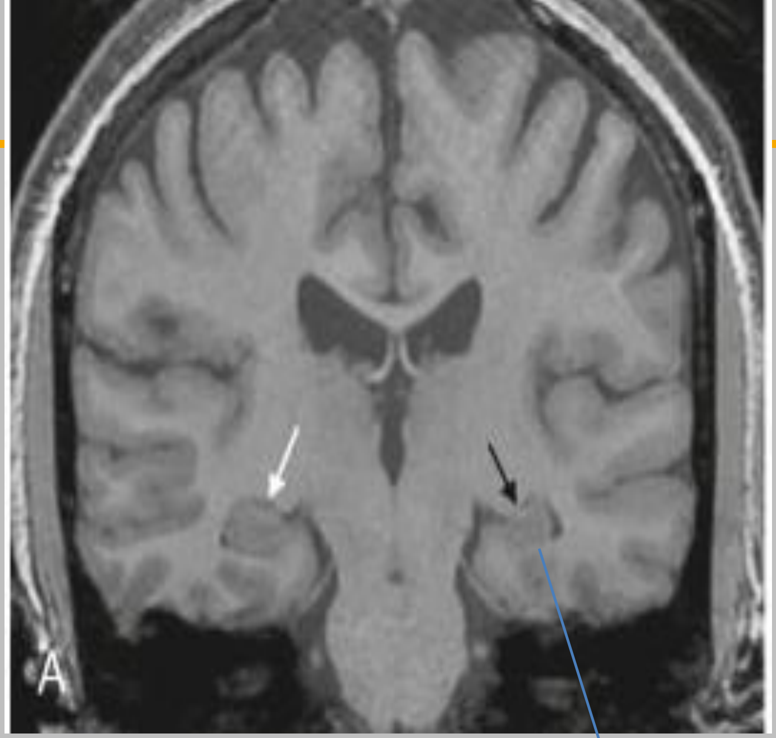
Brain Hamartomas



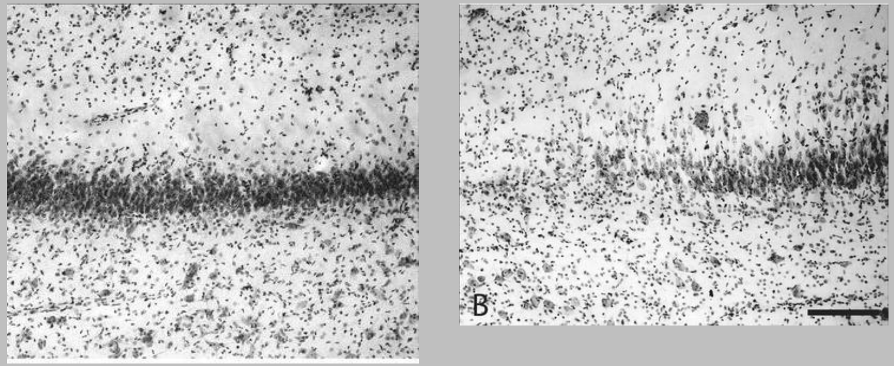

Zitelli & Davis' Atlas of Pediatric Physical Diagnosis, 7th ed.

# Medial Temporal Lobe Epilepsy

- About 30% cases are intractable. Minimal Pharmacological treatment
- Temporal lobe epilepsy accounts for 70% of intractable epilepsy
- MTLE usually manifest as Generalized Tonic-clonic seizure. Starts most commonly in hippocampus (MTL)
- MTL Sclerosis = **scarring** and **volume loss (atrophy)** of medial temporal structures (most notably **hippocampus**)
- Dx: **MRI**
- Characterized by neuronal loss and gliosis of the hippocampus and adjacent structures
- Unclear Pathophysiology of MTS



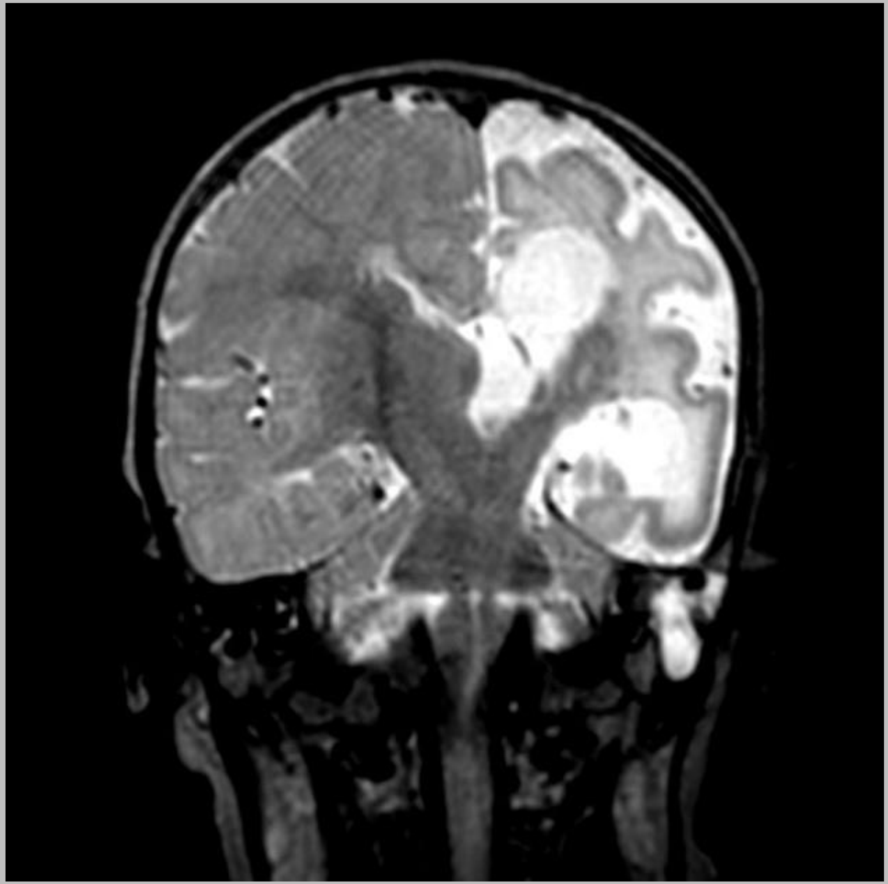

Sclerosis in this hippocampus




Lee TS et al. 2006

# Autoimmune Encephalopathies: Rasmussen Syndrome (RAS-muss-en)

- Rasmussen Syndrome is a rare pediatric encephalitis that effects **one hemisphere of the brain**.
- 1.8 to 2.4 out of every 10 million people annually, or 0.18 per 100,000 people, usually in previously normally developing pediatric boys (aged 6-8 most common)
- Drug resistant seizures are a hallmark feature
- T-cell-dominated encephalitis with **activated microglial cells and reactive astrogliosis is apparent on histopathological examination.** What is causing this microglia activation and subsequent astrogliosis is unknown. No antigen that CNS immune cells attack has been identified
- Characterized by recurrent seizures, delayed developmental milestones, progressive cognitive decline. This can be acquired via through medical history.
- neuroimaging with chronic inflammatory changes, and progressive hemispheric atrophy
- In short, we really don't know what causes this disease
- Treatment involves a hemispherectomy of the affected brain lobe/hemisphere



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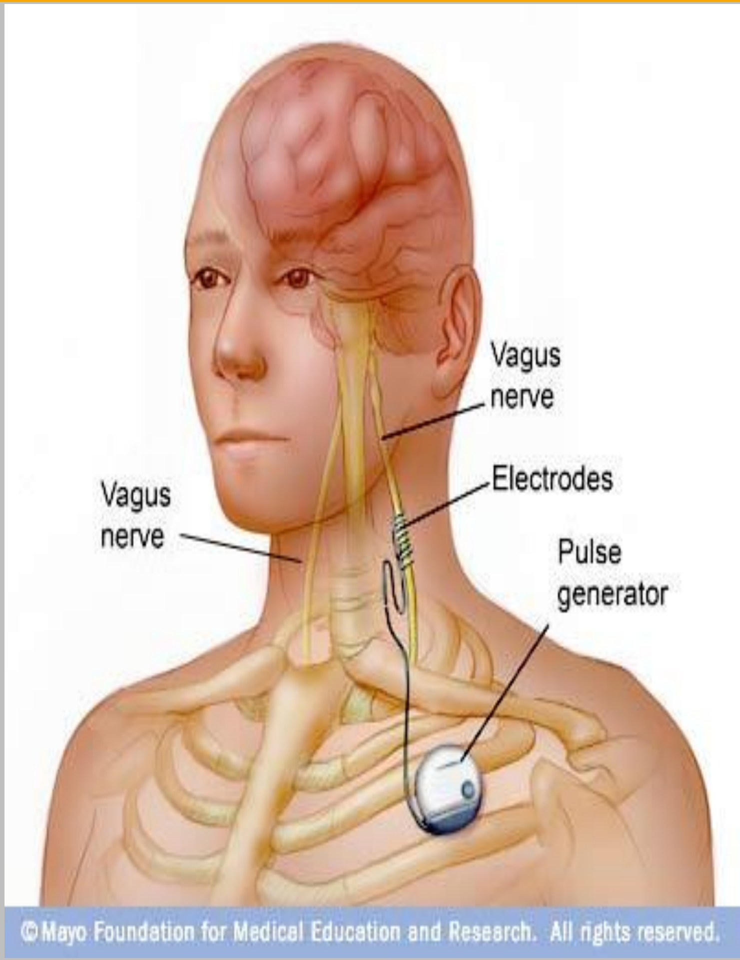
# Autoimmune Encephalopathies: NMDA Encephalitis

- Anti-NMDA Receptor encephalitis is the most common (one affected out of 1.5 million people per year) autoimmune encephalitis
- **IgG antibodies against the NR1 subunit of the NMDA receptors in the central nervous system(CNS)** are a hallmark of this disease. Unlike Rasmussens, we have an idea about what antibody is attacking what antigen. These antibodies are identifiable in the serum or the cerebrospinal fluid (CSF)
- Plasma exchange helps reduce amount of Anti-NMDA receptor antibodies in serum
- Seems to be triggered by viruses such as Herpes simplex or ovarian teratomas (tumor)
- **Transvaginal ultrasonography** is the most crucial test for young women presenting with the illness due to the high incidence of underlying **ovarian teratoma.**
- As with most autoimmune disorders, women aged 25-35 are most commonly affected
- 5 clinical Stages: prodromal phase, psychotic phase, unresponsive phase, hyperkinetic phase, and recovery phase.

1. Prodrome: can mimics common viral infections.
2. Psychotic: Psych symptoms in adults. Includes visual or auditory hallucinations, acute schizoaffective episodes, depression, mania, apathy, anxiety, fluctuating sensorium, bizarre behaviors, hypersexuality, aphasia, amnesia, sleep-wake cycle disruption or eating disorders can appear rapidly within days to weeks in these patients with no prior psychiatric diagnosis. Children who are affected suffer less psych symptoms and more from movement disorders and seizures in this period
3. Unresponsive phase: unresponsive phase characterized by mutism, decreased motor activity, and catatonia
4. Hyperkinetic phase: Lip and tongue smacking common, as well as autonomic instability
5. Recovery of language function and behavioral symptoms occurs at the end.

# Treatments

- Goal of Therapy:
  - Ex: seizure freedom! >> reduction in seizure frequency, no complex seizures, simple partial seizures only
- Pharmacological → future Dr. Goldstein lecture
- Surgery
- Brain stimulation
- Vagus Nerve Stimulator – increase parasympathetic tone
- Ketogenic Diet – not entirely understood mechanism. May effect cellular metabolism enough to increase seizure threshold. Usually diet minimal in carbohydrates, this stimulates lipolysis ketone release from liver (liver cannot use ketones as energy) which may effect resting neuronal membrane potential and overall proclivity to fire action potential
- Stress Modulators
- Yoga
- Acupuncture
- Seizure Alert Dogs

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# Question

- A 75 year-old man with no past medical history presents to the ED with a complaint of abnormal twitching of his right upper arm for the past 2 minutes. The man said he was completely aware he was having a seizure and did not lose consciousness. The patient has no other neurological complaint and has had no neurological issues in the past. After CT and MRI was completed, EEG was positive for seizure activity. Which of the following is true about this patient?
  - A) This patient is having a generalized seizure with seizure activity in the motor cortex bilaterally
  - B) This patient is having a focal impaired awareness seizure in the motor cortex bilaterally
  - C) This patient is having a focal aware seizure in the left motor cortex
  - D) This patient is having a focal aware seizure in the right motor cortex
  - E) This patient is in status epilepticus and needs lorazepam

# Question

A 7-year-old boy with previous normal neuropsychiatric development presents with to the family med physician after being discharged 2 weeks ago from the hospital for a tonic-clonic seizure. About 6 months ago, the mother states her child is  having trouble following commands and listening. Now the patient's mother states her child is not speaking as well as he previously did as is. EEG shows abnormalities in what correlates to Broca's and Wernicke's areas. What is the Diagnosis ad the gene implicated in the pathogenesis of the diagnosis?

A) Absence Seizures, **Cacna1H**
B) EAST Syndrome, ***KCNJ10***
C) Sturge Weber Syndrome, **GNAQ**
D) Autosomal dominant partial epilepsy with auditory features, **RELN**
E) Lennox Gastaut syndrome, **GABRB3**